(MATH200) Differential Equations

Mathematical Concepts

Jungwoo Kim
Department of Mathematics, Pohang University of Science and Technology

---

Contents

0 Notation and Conventions

General mathematical notation and writing follow *(MATH000) Mathematical Language*. The conventions below record only notation specific to differential equations used in this book. Widely used standard alternatives may be listed in preferred order; the first form is used throughout.

Differential Equations

The Fourier transform uses the normalization

$$\widehat{f}(\boldsymbol{\xi}) := \int_{\mathbb{R}^n} f(\mathbf{x})e^{-2\pi i\mathbf{x}\cdot\boldsymbol{\xi}}\,\mathrm{d}\mathbf{x}, \qquad f(\mathbf{x}) = \int_{\mathbb{R}^n} \widehat{f}(\boldsymbol{\xi})e^{2\pi i\mathbf{x}\cdot\boldsymbol{\xi}}\,\mathrm{d}\boldsymbol{\xi},$$

whenever the Fourier inversion formula applies.

Notation	Meaning
y'	First derivative with respect to the understood independent variable.
y''	Second derivative.
$\dot{x}$	First derivative of x with respect to time.
$\ddot{x}$	Second derivative of x with respect to time.
$y_h,\ y_c$	Homogeneous or complementary part of a solution.
y_p	Particular solution.
$c_1, c_2, \ldots$	Arbitrary constants in a general solution.
$W(y_1, \ldots, y_n)(t)$, $W[y_1, \ldots, y_n](t)$	Wronskian of $y_1, \ldots, y_n$.
$\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{g}(t)$, $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{g}(t)$	Linear first-order system in state-vector form.
$\boldsymbol{\Phi}(t)$	Fundamental matrix of a linear system.
$\mathcal{L}\{f(t)\}(s)$, $\mathcal{L}[f](s)$	Laplace transform of f.
$\mathcal{L}^{-1}\{F(s)\}(t)$, $\mathcal{L}^{-1}[F](t)$	Inverse Laplace transform of F.
$(f * g)(t)$	One-sided convolution, $(f * g)(t) := \int_0^t f(\tau)g(t-\tau)\,\mathrm{d}\tau$.
$H(t-a),\ u(t-a)$	Heaviside or unit-step function shifted to $t = a$; its value at the jump is specified when relevant.
$G(x, \xi)$	Green's function; the variables depend on context.
$J_\nu(x)$	Bessel function of the first kind of order ν.
$Y_\nu(x),\ N_\nu(x)$	Bessel function of the second kind (Neumann function) of order ν.
$\widehat{f}(\boldsymbol{\xi}),\ \mathcal{F}f(\boldsymbol{\xi})$	Fourier transform under the normalization above.

1 (MATH200) Differential Equations